Abdul Khurram

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Education & Awards

University of British Columbia — Vancouver, BC, Canada
*Bachelor of Science in Computer Science (**Expected Graduation: May 2027**)*

Euclid Mathematics Contest, University of Waterloo — 2021
Top 5% Internationally, Certificate of Distinction

Work Experience

RBC Royal Bank | *Python, Java/Scala, Apache Spark, Linux* — Jan. 2026 – Present
Software Developer Co-op — *Vancouver, BC, Canada*

- Designed and shipped Python/Java services that process **50M+ daily transactions** against petabyte-scale data, modeling complex entity relationships across multiple financial product lines.
- Profiled Spark batch jobs, identified serialization and partition-skew bottlenecks, and re-wrote critical paths with Spark-native APIs and adaptive query tuning — cutting end-to-end **latency by 40%**.
- Architected idempotent, exactly-once batch pipelines with deterministic retry semantics and automated failover, ensuring correctness guarantees across distributed banking workloads.

Contello.ai | *Python, TypeScript/Node.js, AWS* — Jan. – Dec. 2025
Software Engineer Intern — *Toronto, ON, Canada*

- Designed async processing pipelines (SQS, S3) handling **thousands of jobs/day** — decoupled producers from consumers, implemented dead-letter queues, and reasoned about ordering and delivery guarantees.
- Built end-to-end test infrastructure (E2E + integration) across containerized service stacks, gating every release through CI and catching failure modes before production.
- Raised service reliability to **99.9%** by analyzing failure distributions, adding structured retry with exponential backoff, and wiring alerting on latency percentiles and error budgets.

Research Experience

UBC Emerging Media Lab | *C++, Python, Unreal Engine* — Sept. – Dec. 2024
Software Developer & Team Lead — *Vancouver, BC, Canada*

- Led an interdisciplinary research team (CS × Nursing) building an AI-driven VR simulation in C++ for clinical training — scoped requirements, designed the system, and managed delivery.
- Designed a multithreaded supervisory architecture coordinating concurrent AI actors with shared state, lock-free message passing, and real-time protocol enforcement.
- Profiled the inference pipeline, identified memory allocation hotspots, and restructured the call graph to achieve **40% latency reduction** and **90% cost savings**.

Projects

Backer | *TypeScript, Python, DynamoDB, XGBoost* — **Hack The Coast 2026**

- Modeled 10+ entity types with complex many-to-many relationships and transactional consistency constraints on a NoSQL store, designing typed repository abstractions over DynamoDB.
- Trained an XGBoost ranking model on 6 engineered features (stage affinity, text similarity, tag overlap, engagement signals) to score and surface relevant startups from event-stream data.

Workbase | *TypeScript, PostgreSQL, AWS Bedrock* — **Personal Project**

- Designed a multi-step structured output pipeline — LLM generation, schema validation, automated repair, and evidence attribution — with deterministic retry and audit logging.
- Wrote test suites covering domain state machines (status transitions, eligibility rules, deduplication logic) and the structured LLM client (parsing, repair, validation).

Technical Skills

Languages: Python, C/C++, Java, Scala, TypeScript/JavaScript, SQL
Tools & Platforms: Apache Spark, PostgreSQL, AWS (Lambda, DynamoDB, S3, SQS), Docker, Git, Linux